

# Ziqian Zhong

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## Education

**Ph.D. in Computer Science**, Carnegie Mellon University 01/2025 – present

Conduct research around AI safety and interpretability to better understand, align and control AI systems. Advised by Aditi Raghunathan.

**B.S. in Computer Science & Mathematics**, Massachusetts Institute of Technology 08/2020 – 06/2024

GPA: 5.0/5.0. Selected Coursework: Quantitative Methods for NLP; Multi-agent Communication; Machine Learning; Advanced Complexity Theory; Fundamentals of Statistics.

## Experiences

**Research Intern**, Translucent 05/2026 – 08/2026

Conducted and published research on situation awareness with Cassidy Laidlaw and Jacob Steinhardt.

**Research Scientist**, Pika Labs / Mellis Inc. 06/2024 – 01/2025

Developed and improved large-scale video generation models. Contributed to Pika 1.5 and led the release of Pika 2.0.

**Deep Learning Undergraduate Researcher**, MIT 08/2022 – 06/2024

Worked on interpreting and understanding neural networks with Neil Thompson, Jacob Andreas, Max Tegmark, and Ziming Liu.

**Theoretical Computer Science Undergraduate Researcher**, MIT 10/2021 – 05/2022

Member of research team guided by Virginia Vassilevska Williams. Co-discovered new results in graph theory and combinatorics, leading to several publications.

**Algo Developer Intern**, Hudson River Trading 05/2023 – 08/2023

Conducted market and algorithmic research; project featured in [intern spotlights](#).

## Invited Talks

- **Two Stories in Mechanistic Explanation of Neural Networks**. NeurIPS 2023 Oral with Ziming Liu.
- **New Approach for Unbounded SubsetSum**. SODA 2023.
- **Let's talk about reward hacking**. MIT AI alignment (MAIA), 2025.
- **Can we fix reward hacking?** Carnegie Mellon AI Safety Initiative (CASI), 2026.
- **How can we stop reward seekers?** Jane Street Research Symposium, 2026.

## Selected Works

- **User Awareness in Frontier Models**. Ziqian Zhong, Aditi Raghunathan, Cassidy Laidlaw, Jacob Steinhardt. Translucent Blog, August 2026 ([Link](#)).
- **Hardening Agent Benchmarks with Adversarial Hacker-Fixer Loops**. Ziqian Zhong, Ivgeni Segal, Ivan Bercovich, Shashwat Saxena, Kexun Zhang, Aditi Raghunathan. Preprint, 2026 ([Link](#)).

- **Hodoscope: Unsupervised Monitoring for AI Misbehaviors.** Ziqian Zhong, Shashwat Saxena, Aditi Raghunathan. Preprint, 2026 ([Link](#)).
- **ImpossibleBench: Measuring LLMs' Propensity of Exploiting Test Cases.** Ziqian Zhong, Aditi Raghunathan, Nicholas Carlini. ICLR 2026 ([Link](#)).
- **Pando: Do Interpretability Methods Work When Models Won't Explain Themselves?.** Ziqian Zhong, Aashiq Muhamed, Mona T. Diab, Virginia Smith, Aditi Raghunathan. COLM 2026 ([Link](#)).
- **Model self-identification could be subliminally transferred.** Ziqian Zhong. LessWrong, July 2026 ([Link](#)).
- **Spurious probes as a black-box alternative to activation probing.** Ziqian Zhong. LessWrong, September 2026 ([Link](#)).
- **Watch the Weights: Unsupervised monitoring and control of fine-tuned LLMs.** Ziqian Zhong, Aditi Raghunathan. ICLR 2026 ([Link](#)).
- **Base Models Look Human To AI Detectors.** Yixuan Even Xu, Ziqian Zhong, Aditi Raghunathan, Fei Fang, J. Zico Kolter. Preprint, 2026 ([Link](#)).
- **Early Data Exposure Improves Robustness to Subsequent Fine-Tuning.** Lawrence Feng, Gaurav R. Ghosal, Jacob Mitchell Springer, Ziqian Zhong, Aditi Raghunathan. COLM 2026 ([Link](#)).
- **Algorithmic Capabilities of Random Transformers.** Ziqian Zhong, Jacob Andreas. NeurIPS 2024 ([Link](#)).
- **The Clock and the Pizza: Two Stories in Mechanistic Explanation of Neural Networks.** Ziqian Zhong\*, Ziming Liu\*, Max Tegmark, Jacob Andreas. NeurIPS 2023 (Oral) ([Link](#)).
- **Grokking as Compression: A Nonlinear Complexity Perspective.** Ziming Liu, Ziqian Zhong, Max Tegmark. NeurIPS 2023 UniReps Workshop ([Link](#)).
- **On Problems Related to Unbounded SubsetSum: A Unified Combinatorial Approach.** Mingyang Deng\*, Xiao Mao\*, Ziqian Zhong\*. SODA 2023 ([Link](#)).

## Selected Awards

- Gold Medal, 4<sup>th</sup> Place – International Olympiad in Informatics 2019
- First Place – Meta Hackercup 2024
- Gold Medal – 46<sup>th</sup> ICPC World Final
- Honorable Mention – Alibaba Global Mathematics Competition 2022
- Honorable Mention – Putnam Mathematical Competition 2022

## Selected Projects

- **Is my problem new?** ([Link](#), 11/2023). Employ LLMs and vector embeddings to detect similarity among competitive programming problems. ~40k monthly page views and adopted by major contest setters.
- **CP Ideas** ([Link](#), 07/2022). Generates competitive programming problems via fine-tuned GPT-3. Dataset collected and cleaned from various online judges.